

Of Rock Rabbits and Elon Musk

Martin Kunz (University of Geneva)

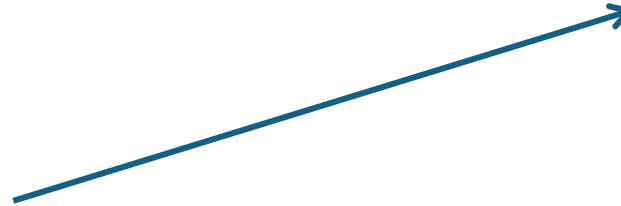
or really:

Chris Finlay (EPFL) and **Ajith Sampath** (UniGe)

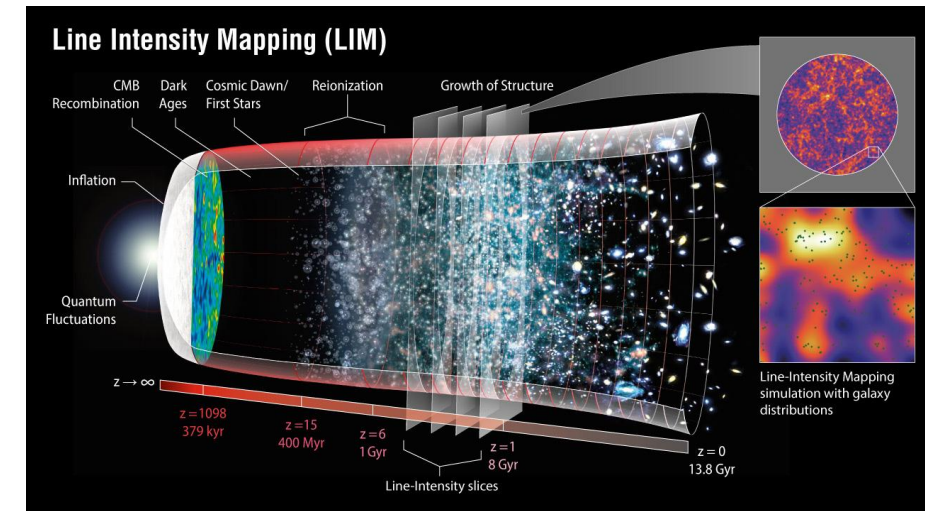
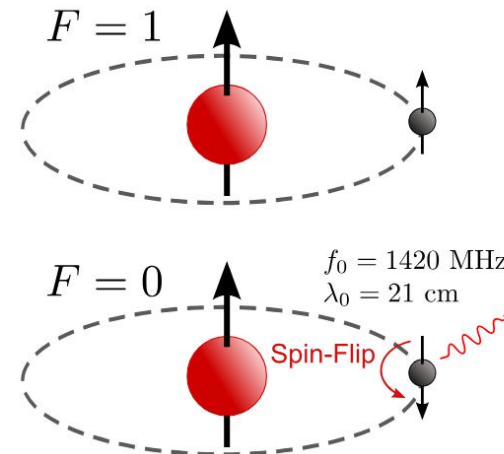
and the HIRAX and TABASCAL teams



From Planck to Rock Rabbits



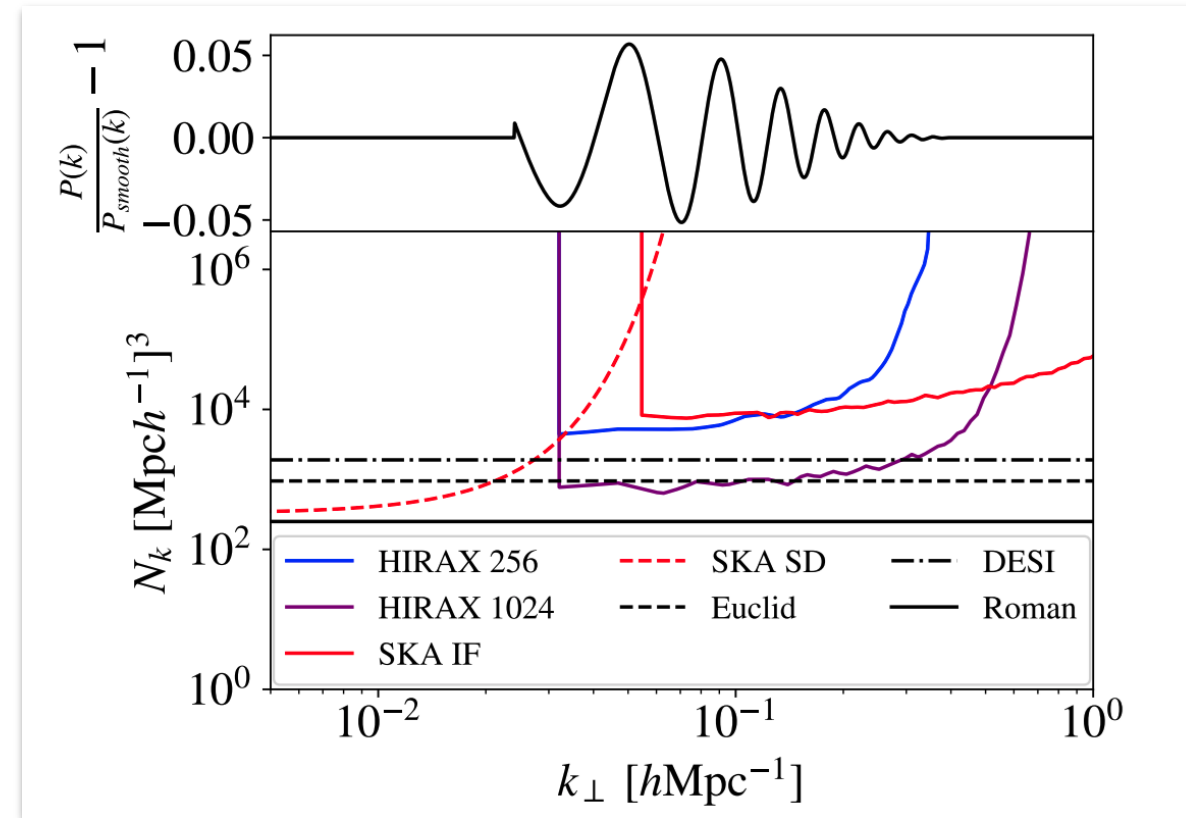
Credits: Cynthia Chiang



Credits: NASA/Lambda

What is the **H**ydrogen **I**ntensity and **R**ead-time **A**nalysis **e**Xperiment ?

- Redshift range : $0.8 < z < 2.5$ to capture DE domination at $z \sim 1$ and for sufficient volume
→ **Frequencies 400 MHz - 800 MHz**
- Sensitive to medium to large scales:
BAO: 150 Mpc : 3 - 1.3 deg
→ **15m - 60m → compact array**
- BAO radial scale : 20 to 12 MHz resolution
→ **At least 100 channels (more for foregrounds and higher-order peaks)**
- BAO signal level : ~ 0.1 mK
→ **low T_{sys} and large collecting area (many dishes)**



D. Crichton et al. 2021

→ 32x32 compact (and redundant) array of 6m dishes (scalable $8 \rightarrow 128 \rightarrow 256 \rightarrow 1024$),
1024 frequency channels, drift-scanning with manual repointing, much CHIME/CHORD heritage

UNIVERSITY OF KWAZULU-NATAL
INKYUVESI
YAKWAZULU-NATALI

BIOT
BIOMEDICAL INSTITUTE OF TECHNOLOGY

UNIVERSITY OF BRITISH COLUMBIA

UNIVERSITÄT SÄKTZ
UNIVERSITY

EPFL

ETH zürich

West Virginia University

UNIVERSITY OF OXFORD

UNIVERSITY OF CAPE TOWN
UNIVERSITY OF CAPE TOWN

RHODES UNIVERSITY
Where leaders learn

WISCONSIN
UNIVERSITY OF WISCONSIN

UNIVERSITÉ DE GENÈVE

DUT DURBAN
UNIVERSITY OF TECHNOLOGY

AIMS

SARAO
South African Radio Astronomy Observatory

NRF
National Research Foundation

JPL
Jet Propulsion Laboratory
California Institute of Technology

McGill
UNIVERSITY

SIM NS
FOUNDATION

Yale



Dish Transport



Credits : Anthony Walters

Dish Transport



Credits : Anthony Walters

Dish Installation



Credits : Anthony Walters

HIRAX current status



(All picture credits: Jennifer Studer & Antony Walters)

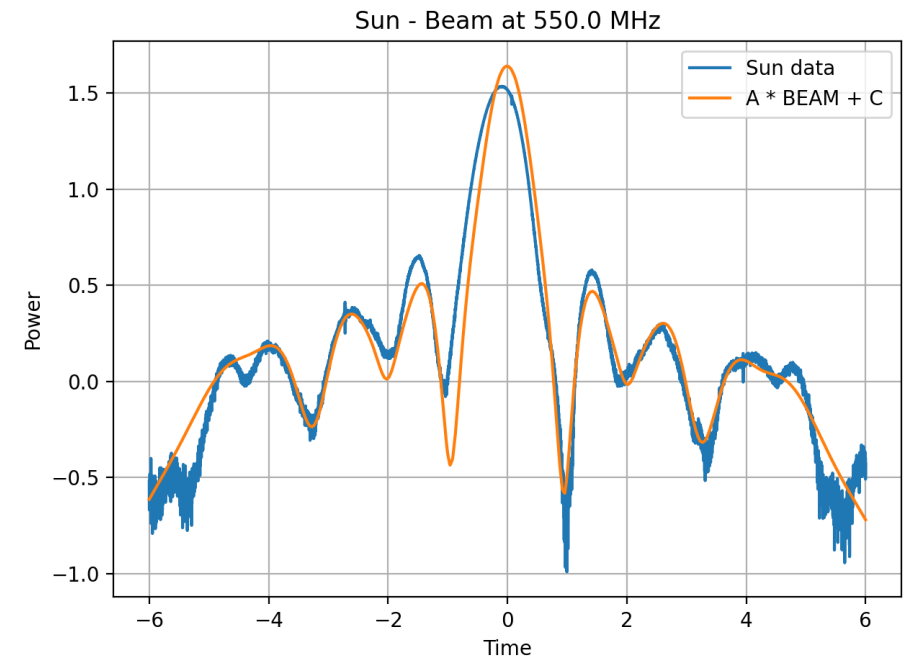
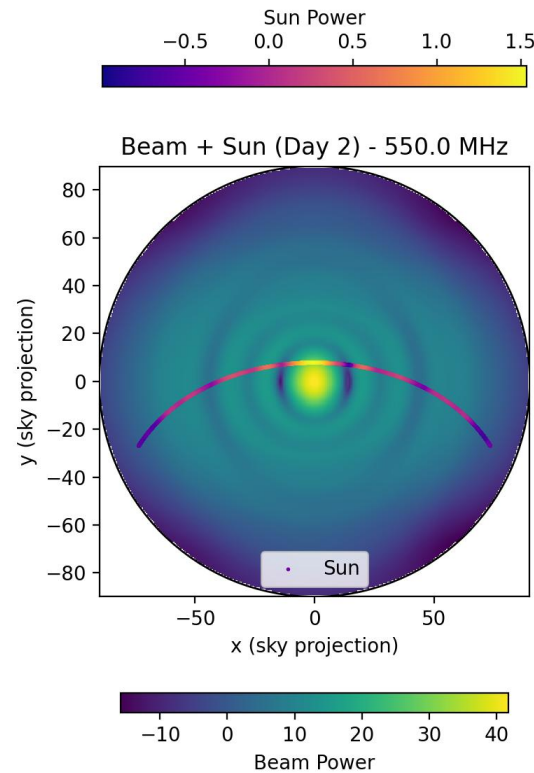
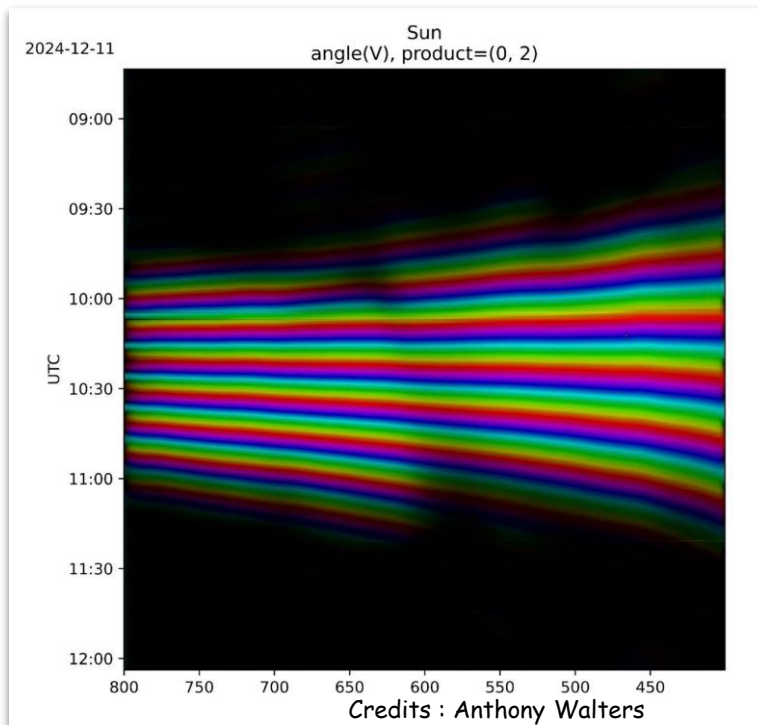


HIRAX 'first fring

Beams are very important for HI intensity mapping as foregrounds are huge and will leak into the signal through side-lobes



Measured vs simulated HIRAX beams:

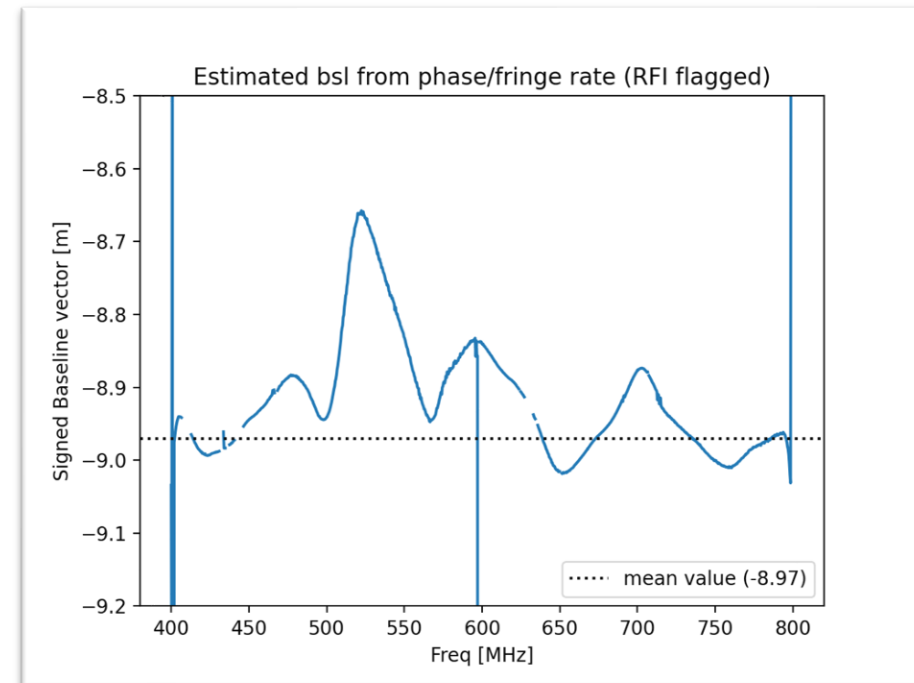
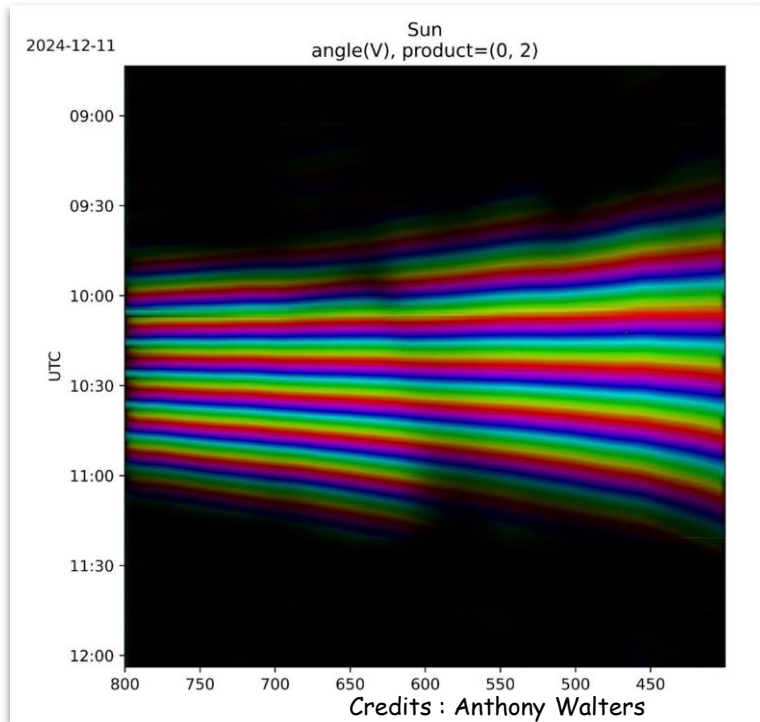


Analysis & figures by Ajith Sampath

Help for forgetful cosmologists



What do you do when your array is far away in the desert and you cannot remember the dish spacing ?!



Analysis & figure by Ajith Sampath

The SKA South Africa site

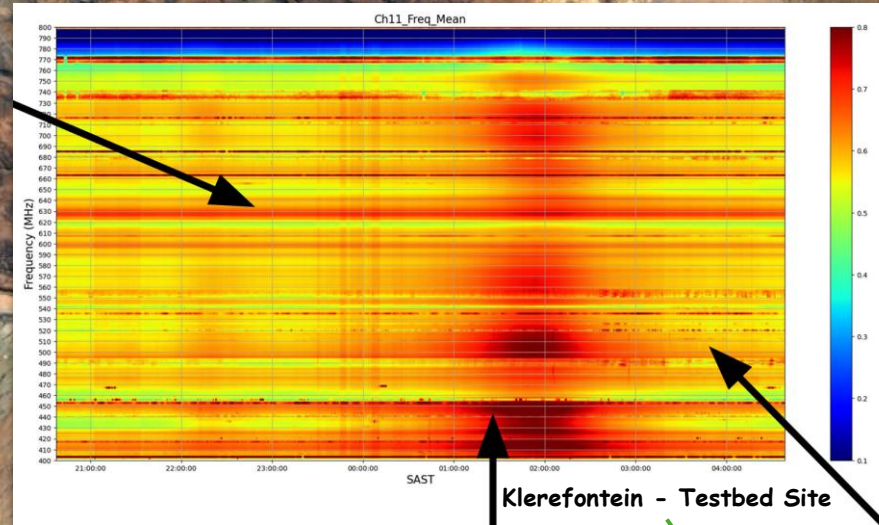
Bergsig

HIRAX

Corrupt channels

Klerkshoop

Onverwags



Kloof

Saaipoort

Time varying
RFI

Galactic
transit

Blinkklip

Asbosbam

R361

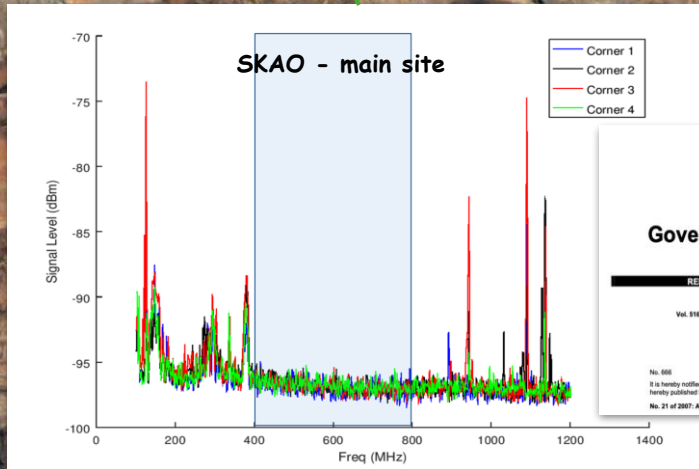
R386

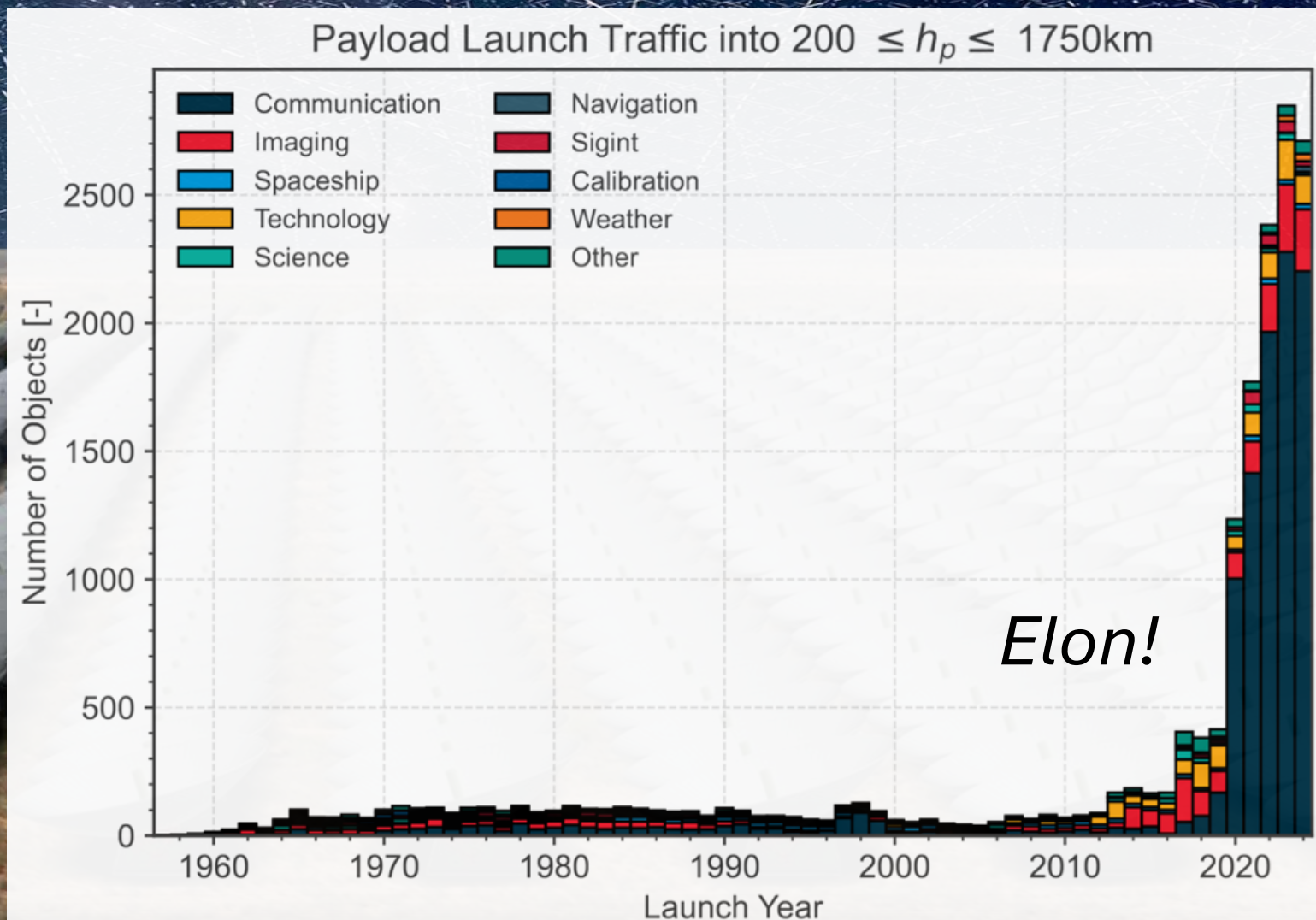
R384

R63

Carnarvon

R63





RASCAL? No, TABASCAL!

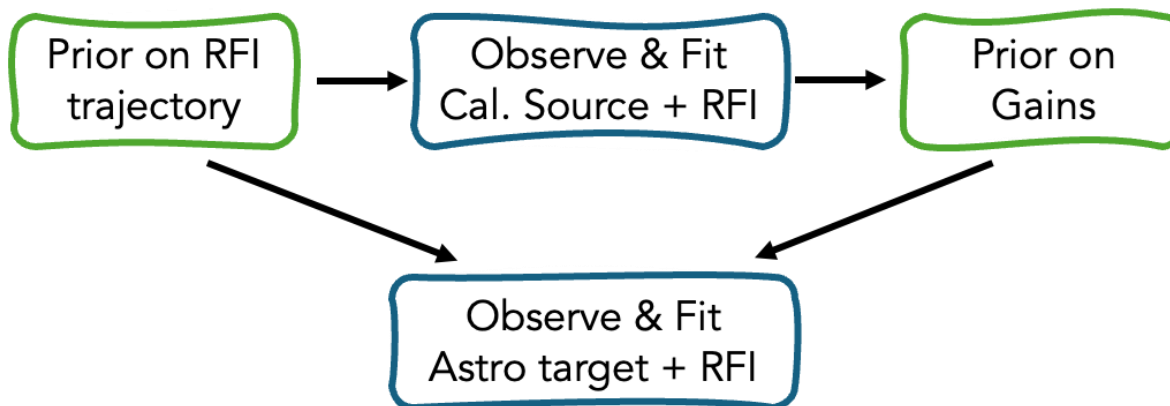


A very relaxed looking
Chris Finlay

In many frequency bands, 1/4 – 1/3 of radio data is flagged because of RFI, often satellites!

TABASCAL - [arXiv:2301.04188 \(MNRAS 2023\)](#), [arXiv:2502.00106 \(A&A 2025\)](#)

- Bayesian forward model for RFI signal, astronomical signal, and antenna gains
- Use RFI trajectories to separate astronomical and RFI signals
- Originally **Chris** w/ Bruce Bassett, Nadeem Oozer and me
- EPFL project to move to TIER-0 GPU HPC supercomputers (w/ Emma Tolley) making good progress

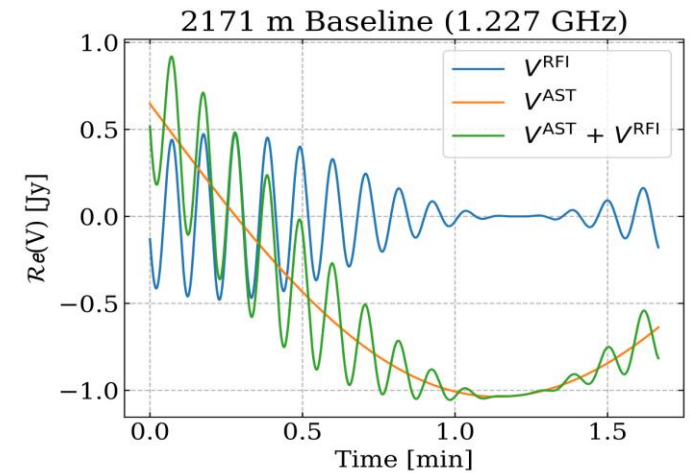


The Reverend says hello

Principles:

- Satellites move much faster than the sky.
 - Their trajectories are (mostly) predictable from so-called TLE's.
 - Interferometers provide a lot of information for self-calibration.
- We can forward-model the whole telescope + sky + satellites system and sample / optimize the posterior in a Bayesian framework.

(We can even calibrate off the RFI! ☺)



8 Chris Finlay et al.

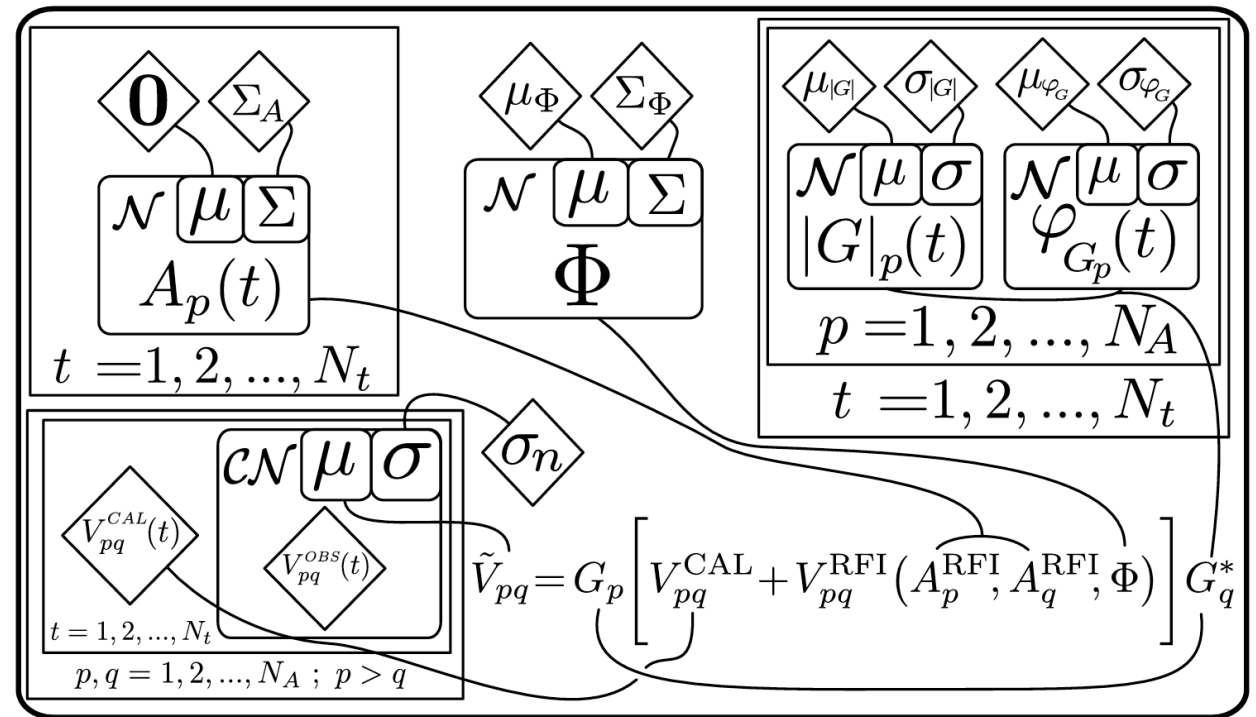
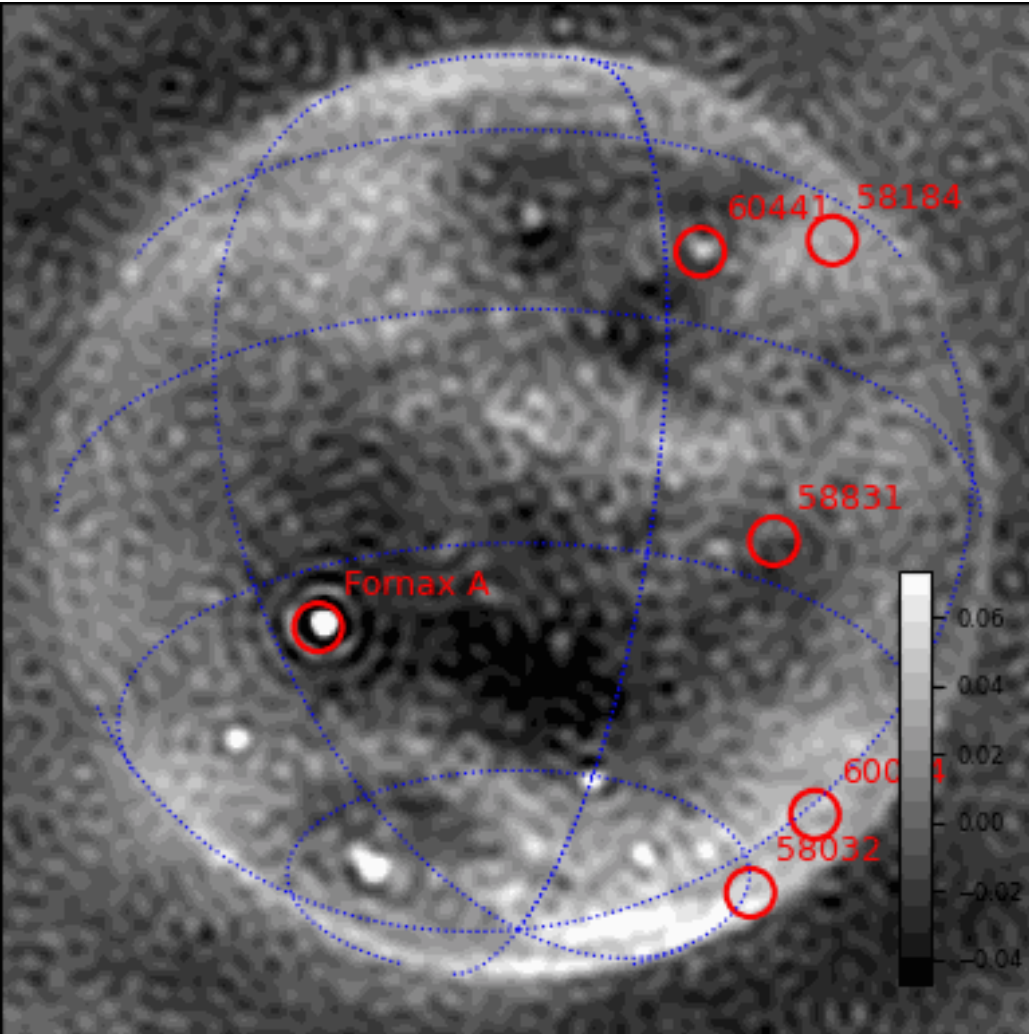


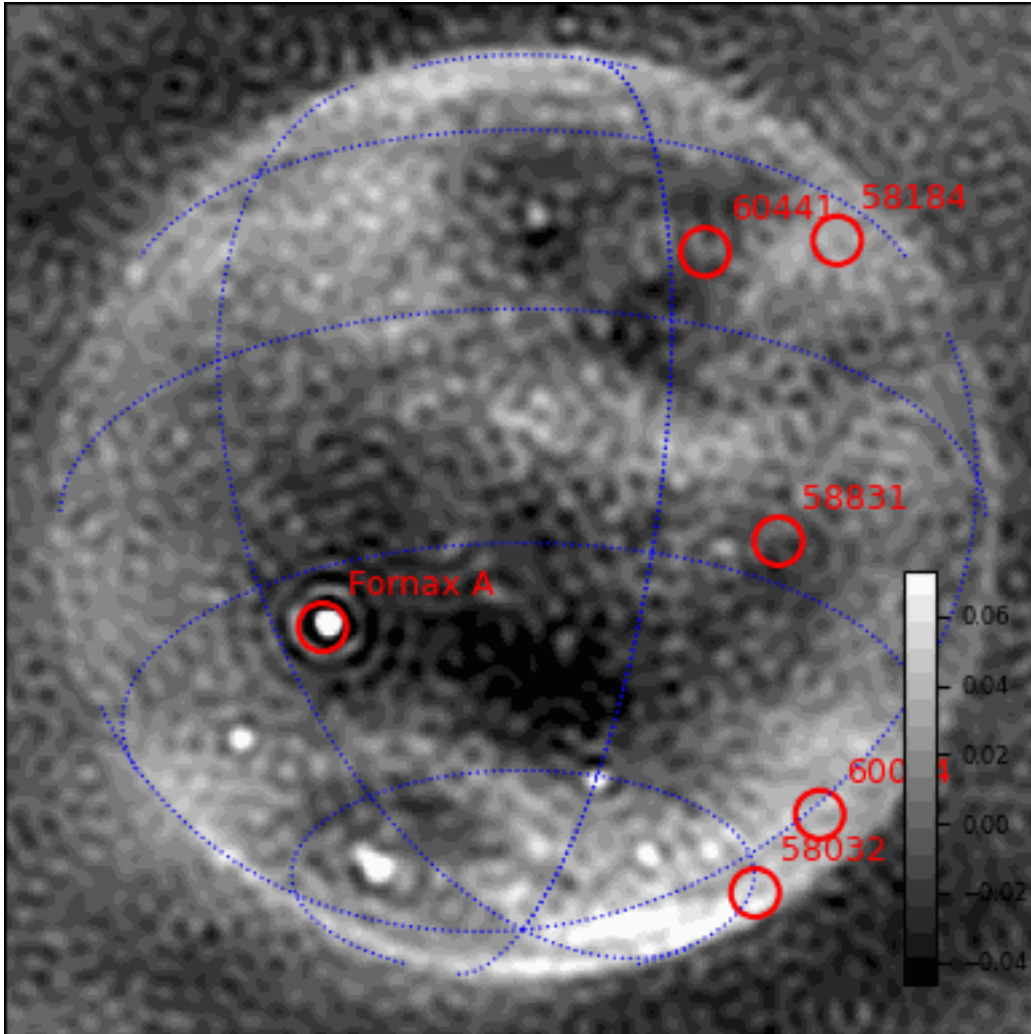
Figure 2. A Bayesian factor graph of the probabilistic model used to estimate uncalibrated and RFI contaminated visibilities. The constants are shown as diamonds. The free parameters of the model are shown as rectangles with rounded corners. Their distributions are shown in the top left corner with the parameterization of the distribution given in the smaller rounded rectangles in the top right corner. Repeated parameters that are indexed are placed in a rectangle, with sharp corners, known as a plate with the index repetition indicated at the bottom centre of the rectangle. The rectangles in the top half of the diagram form the prior over the parameters and the rectangle in the lower left is the likelihood term. The equation in the lower right of the diagram is the mathematical model for the observed data.

Starlink V2-Mini Observation - 144 s, 170.3 MHz, 26 Channel

Raw Data

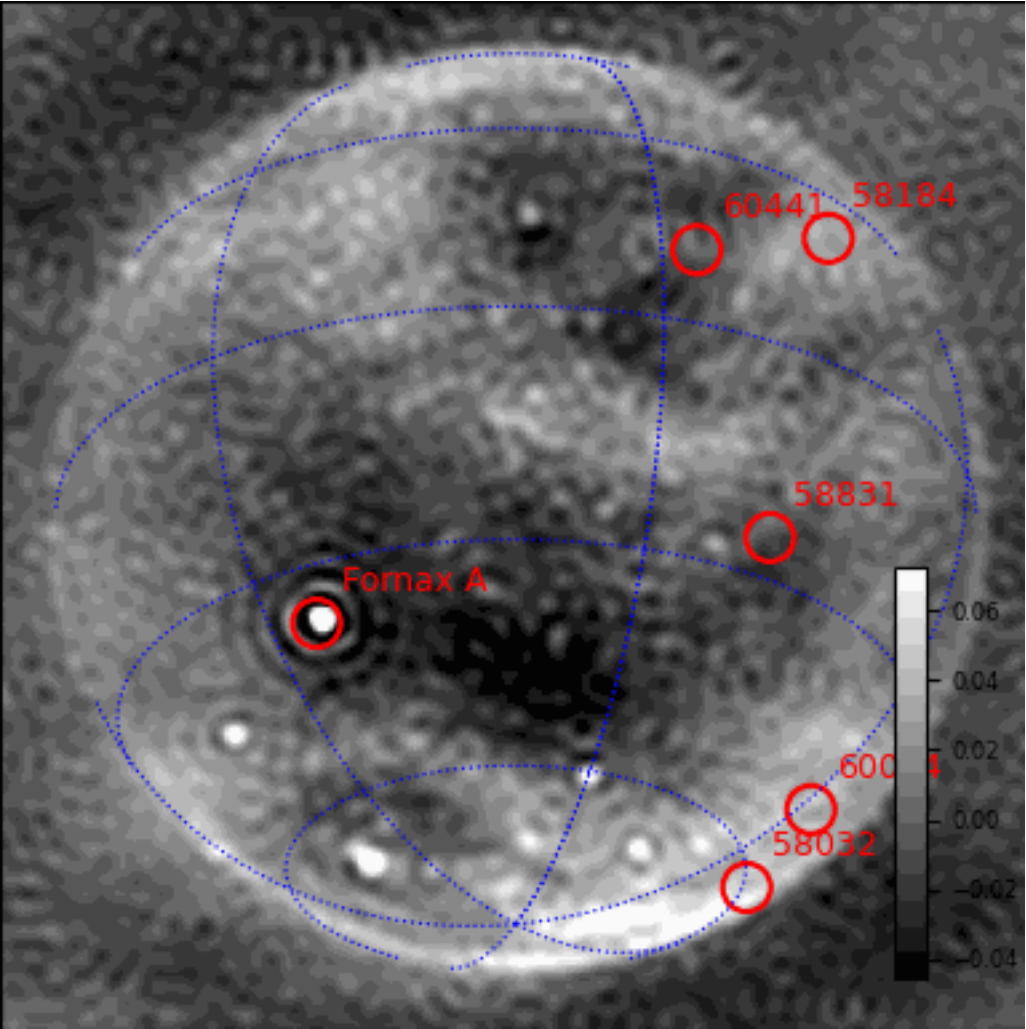


RFI Subtracted Data



Starlink V2-Mini Observation - 144 s, 170.3 MHz, 26 Channel

Astronomical Estimate



RFI Estimate

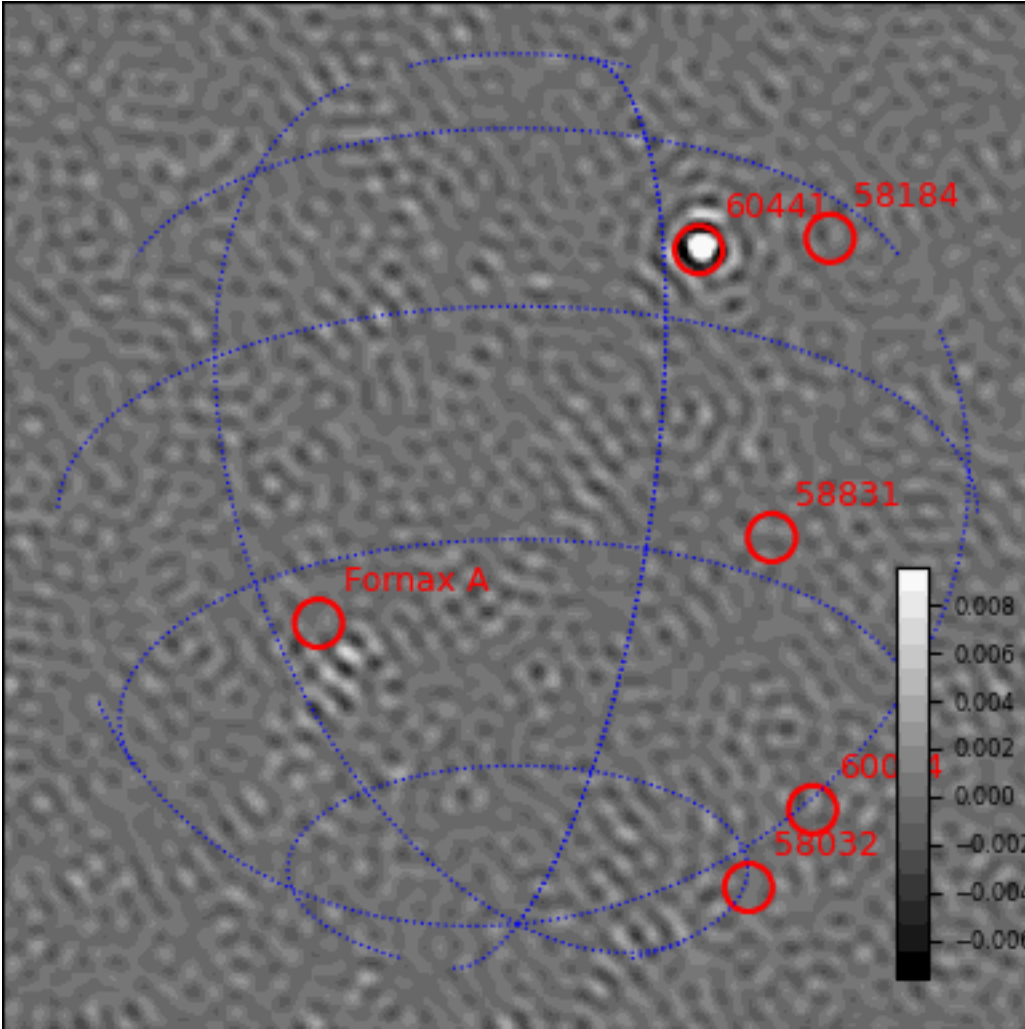


Image credit: Chris Finlay
PASC project w/ TABASCAL, E. Tolley, S.J. Tingay and D. Grigg

The circle closes

- ***Andrew likes the CMB***
 - Line intensity mapping is like having many CMBs
- ***Andrew likes statistics***
 - There are numerous statistical opportunities in these observations
 - We are now working with Andrew, Alan and others on TABASCAL AI/SBI
- ***We like Andrew***
 - Happy Birthday, Andrew!

